Arista Access Node MAR-ACC-04 MARATHAHALLI CORE Hardware/Environmental Fault Resolution SOP

Node: MAR-ACC-04 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1078

Description

The Access Node MAR-ACC-04, located at the Marathahalli Core site, equipped with Arista vendor equipment in the Access role, is susceptible to Hardware/Environmental Faults resulting in ASIC memory parity errors, line card crashes, and power supply redundancy loss. This failure domain occurs due to the intricate interaction of hardware components and environmental factors. The Arista 7150S-64C switch, a high-performance, 10GbE access switch, employs a modular architecture with redundant power supplies, line cards, and ASICs to ensure maximum uptime and reliability. However, in the event of a hardware or environmental failure, the switch's control and data planes can become destabilized, leading to alarm conditions such as ALARM_1078. This SOP outlines the procedures for diagnosing and resolving such failures to restore the Access Node to a stable and operational state.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the Marathahalli Core site.

- Check the status of the redundant links connecting the Access Node to the core network.

- Ensure that all necessary safety protocols are in place to prevent damage to equipment or injury to personnel.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: show system redundancy, Expected Output: Redundancy status: Active, Description: This command verifies the redundancy status of the system, including the power supplies and line cards. A 'Redundancy status: Active' output indicates that the system is properly configured for redundancy, while a 'Redundancy status: Non-Active' output may indicate a potential hardware or environmental failure.

Command: show system asic, Expected Output: ASIC status: Online, Description: This command displays the status of the ASIC, including its online status and any error messages. A 'ASIC status: Online'
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output indicates that the ASIC is functioning properly, while a 'ASIC status: Offline' output may indicate a potential hardware failure.

Command: show system power, Expected Output: Power status: Redundant, Description: This command verifies the power status of the system, including the power supplies and their redundancy status. A 'Power status: Redundant' output indicates that the system is properly configured for power redundancy, while a 'Power status: Non-Redundant' output may indicate a potential hardware or environmental failure.
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Resolution Steps

Step 1: Step: Replace faulty power supply, Command: no power supply 1, Description: Replace the faulty power supply with a new one, and then configure the system to use the new power supply.

Step 2: Step: Replace faulty line card, Command: no line card 1, Description: Replace the faulty line card with a new one, and then configure the system to use the new line card.

Step 3: Step: Reset ASIC, Command: reset asic, Description: Reset the ASIC to its default configuration, and then reconfigure the system to its previous state.

Post-Check Verification

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Command: show system redundancy, Expected Output: Redundancy status: Active, Description: Verify that the system redundancy status is active, indicating that the system is properly configured for redundancy.

Command: show system asic, Expected Output: ASIC status: Online, Description: Verify that the ASIC status is online, indicating that the ASIC is functioning properly.

Command: show system power, Expected Output: Power status: Redundant, Description: Verify that the power status is redundant, indicating that the system is properly configured for power redundancy.
```

Roll-back Procedures

- Step: Restore original configuration, Command: load configuration /bootflash/config.txt, Description: Restore the original configuration from the backup file, and then verify that the system is functioning properly.

- Step: Reboot system, Command: reload, Description: Reboot the system to ensure that all changes are applied, and then verify that the system is functioning properly.